

Technical Description SIGMA CONTROL 2

Process Map $\geq$ dry_3.1.1

7_7601_DRY_PA_06E

General

1. In communication mode "send/receive" Sigma Control can monitor the master program. This takes place with "usiSlaveMsgHeader" and "usiSlaveMsgFooter" that have to be returned unchanged from master.

<u>Example sequence for Profibus Master in Step 7 :</u>	<u>Comment</u>
L DB (receive box master).DBW 94	Message header
T DB (transmit box master).DBW 38	
L DB (receive box master).DBW 126	Message footer
T DB (transmit box master).DBW 62	

Monitoring can be activated in menu *"Communication / Com-Module / / Communication error: Timeout"* by entering a check mark at check box right of Timeout.
If monitoring is not required the above sequence is not used.

2. Further information on signal use can be obtained from Kaeser Service Support Engineering, phone ++49 9561 640-7979

Sigma Control 2 Profibus

1. The data required for planning a Profibus project are in the equipment specification file "KAES0CEC.gsd".
2. The Sigma Control is a Profibus DPV0 slave with 64 bytes of input data and 128 bytes of output data. Depending on the number of Sigma Control units used the Profibus DP master must have a correspondingly large enough address space.

Sigma Control 2 Profinet

1. The data required for planning a Profinet project are in the equipment specification file "GSDML-V2.25-Kaeser-SC2-20120203.xml".

Sigma Control 2 Modbus**1. Compressor ==>>> Master Controller**

Read Input Registers with modbus function 04 "Read Input Registers" or read Holding Registers with modbus function 03 "Read Holding Registers".
Modbus Protocol address: Address of the data at the bus (Base 0).
Modbus Register: Register address of the data = PLC-address (Base 1).
3xxxx 3 = Identification of Input Register, xxxx = Register address
4xxxx 4 = Identification of Holding Register, xxxx = Register address

2. Master Controller ==>>> Compressor

Write registers with function 16 "Write Multiple Registers" or 06 "Write Single Register".
Read registers with function 03 "Read Holding Registers". Never written registers contain "0".
Modbus Protocol address: Address of the data at the bus (Base 0).
Modbus Register: Register address of the data = PLC-address (Base 1).
4xxxx 4 = Identification of Holding Register, xxxx = Register address

Sigma Control 2 Devicenet

1. The data required for planning a DeviceNet project are in the equipment specification file "324-8172-EDS_ABCC_DEV_V_2_3_1.eds".
2. 8 Bit register: Low bytes are shown at low address.

Sigma Control 2 EtherNet/IP

1. The data required for planning an EtherNet/IP project are in the equipment specification file "SC2_EtherNetIP.eds".
2. 8 Bit register: Low bytes are shown at low address.

SAM Send/Receive

1. Data for compressor 1 to 16 can be found in SAM data block 64 to 79.

SIGMA CONTROL 2 Process Map

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Table 1 Process

1. Compressor ==>>>> Master Controller

Table 1: Process data

* adjustable at SC 2 display

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

Sigma Control 2								SAM		Description	Unit	min	max	Format	
JSON-RPC	Profibus/-net		Modbus**				DeviceNet	Send/Receive							
BDSnd	S7 DBW	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	EtherNet/IP Register MSB/LSB	S5 DW	S7 DBW						
	0	0	21	30022	277	40278	1/0	32	64	Alarm xAM16 - xAM1	Table 1.1	binary			BOOL
	2	1	22	30023	278	40279	3/2	33	66	Alarm xAM32 - xAM17	Table 1.1	binary			BOOL
	4	2	23	30024	279	40280	5/4	34	68	Alarm xAM48 - xAM33	Table 1.1	binary			BOOL
	6	3	24	30025	280	40281	7/6	35	70	Alarm xAM64 - xAM49	Table 1.1	binary			BOOL
	8	4	25	30026	281	40282	9/8	36	72	Alarm xAM80 - xAM65	Table 1.1	binary			BOOL
	10	5	26	30027	282	40283	11/10	37	74	Warning xWM16 - xWM1	Table 1.2	binary			BOOL
	12	6	27	30028	283	40284	13/12	38	76	Warning xWM32 - xWM17	Table 1.2	binary			BOOL
	14	7	28	30029	284	40285	15/14	39	78	Warning xWM48 - xWM33	Table 1.2	binary			BOOL
	16	8	29	30030	285	40286	17/16	40	80	Warning xWM64 - xWM49	Table 1.2	binary			BOOL
	18	9	30	30031	286	40287	19/18	41	82	Warning xWM80 - xWM65	Table 1.2	binary			BOOL
	20	10	31	30032	287	40288	21/20	42	84	Operational message xOM16 - xOM1	Table 1.3	binary			BOOL
	22	11	32	30033	288	40289	23/22	43	86	Operational message xOM32 - xOM17	Table 1.3	binary			BOOL
	24	12	33	30034	289	40290	25/24	44	88	Operational message xOM40 - xOM33	Table 1.3	binary			BOOL
	26	13	34	30035	290	40291	27/26	45	90	Binary signal xBS15 - xBS0	Table 1.4	binary			BOOL
	28	14	35	30036	291	40292	29/28	46	92	Binary signal xBS31 - xBS16	Table 1.4	binary			BOOL
	30	15	36	30037	292	40293	31/30	47	94	Binary signal xBS47 - xBS32	Table 1.4	binary			BOOL
	32	16	37	30038	293	40294	33/32	48	96	Binary signal xBS63 - xBS48	Table 1.4	binary			BOOL
iDV0	34	17	38	30039	294	40295	35/34	49	98	Compressor cooling heat exchanger inlet temp. T10		0,1 °C	-990	+3270	INT
iDV1	36	18	39	30040	295	40296	37/36	50	100	Stage 2: air pressure inlet p3		mbar	-4000	+21000	INT
iDV2	38	19	40	30041	296	40297	39/38	51	102	Stage 2: air pressure outlet p4		mbar	-4000	+21000	INT
iDV3	40	20	41	30042	297	40298	41/40	52	104	Air pressure outlet p100		mbar	-4000	+21000	INT
iDV4	42	21	42	30043	298	40299	43/42	53	106	Oil system: oil pressure inlet lubrication p66		mbar	-4000	+21000	INT
iDV5	44	22	43	30044	299	40300	45/44	54	108	Oil system: oil pressure inlet valve p63		mbar	-4000	+21000	INT
iDV6	46	23	44	30045	300	40301	47/46	55	110	Water temperature outlet heat exchanger stage 1 T11		0,1 °C	-990	+3270	INT
iDV7	48	24	45	30046	301	40302	49/48	56	112	Water temperature outlet heat exchanger stage 2 T12		0,1 °C	-990	+3270	INT
iDV8	50	25	46	30047	302	40303	51/50	57	114	Primary water system: water pressure p14		mbar	-4000	+21000	INT
iDV9	52	26	47	30048	303	40304	53/52	58	116	Software version 1st digit / 2nd digit					BYTE/BYTE
iDV10	54	27	48	30049	304	40305	55/54	59	118	Software version 3rd digit / 4th digit					BYTE/BYTE
iDV11	56	28	49	30050	305	40306	57/56	60	120	AnMod_I_1		*	-1000	+1000	INT
iDV12	58	29	50	30051	306	40307	59/58	61	122	Rotation dryer: Pressure dew point M85		0,1 °C	-800	+200	INT
iDV13	60	30	51	30052	307	40308	61/60	62	124	Pressure actual value pN (pressure regulation)		mbar	-6250	32767	INT
iDV14	62	31	52	30053	308	40309	63/62	63	126	Current drive motor starts/h		1			INT

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Table 1 Process

1. Compressor ==>>>> Master Controller

Table 1: Process data

* adjustable at SC 2 display

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

Sigma Control 2								SAM		Description	Unit	min	max	Format
JSON-RPC	Profibus/-net		Modbus**				DeviceNet	Send/Receive						
BDSnd	S7 DBW	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	Register MSB/LSB	S5 DW	S7 DBW					
iDV15	64	32	53	30054	309	40310	65/64	64	128	Total compressor hours run x 1000	h			INT
iDV16	66	33	54	30055	310	40311	67/66	65	130	Total compressor hours run x 1	h			INT
iDV17	68	34	55	30056	311	40312	69/68	66	132	Hours on load x 1000	h			INT
iDV18	70	35	56	30057	312	40313	71/70	67	134	Hours on load x 1	h			INT
iDV19	72	36	57	30058	313	40314	73/72	68	136	Setpoint pressure pA switching point	mbar			INT
iDV20	74	37	58	30059	314	40315	75/74	69	138	Setpoint pressure pA switching difference	mbar			INT
iDV21	76	38	59	30060	315	40316	77/76	70	140	Setpoint pressure pB switching point	mbar			INT
iDV22	78	39	60	30061	316	40317	79/78	71	142	Setpoint pressure pB switching difference	mbar			INT
iDV23	80	40	61	30062	317	40318	81/80	72	144	Nominal package pressure	mbar			INT
iDV24	82	41	62	30063	318	40319	83/82	73	146	Increased cut-out pressure pE	mbar			INT
iDV25	84	42	63	30064	319	40320	85/84	74	148	Remaining drive motor idle time	s			INT
iDV26	86	43	64	30065	320	40321	87/86	75	150	AnMod_p_1	mbar	-4000	+21000	INT
iDV27	88	44	65	30066	321	40322	89/88	76	152	AnMod_p_2	mbar	-4000	+21000	INT
iDV28	90	45	66	30067	322	40323	91/90	77	154	AnMod_T_1	0,1 °C	-990	+3270	INT
iDV29	92	46	67	30068	323	40324	93/92	78	156	AnMod_T_2	0,1 °C	-990	+3270	INT
usiSlaveMsgHeader	94	47	68	30069	324	40325	95/94	79	158	Message header slave				UINT
iDV30	96	48	69	30070	325	40326	97/96	80	160	Stage 1: air inlet temperature T1	0,1 °C	-990	+3270	INT
iDV31	98	49	70	30071	326	40327	99/98	81	162	Stage 1: air outlet temperature T2	0,1 °C	-990	+3270	INT
iDV32	100	50	71	30072	327	40328	101/100	82	164	Heat exchanger stage 1: air outlet temperature T31	0,1 °C	-990	+3270	INT
iDV33	102	51	72	30073	328	40329	103/102	83	166	Stage 2: air outlet temperature T4	0,1 °C	-990	+3270	INT
iDV34	104	52	73	30074	329	40330	105/104	84	168	Heat exchanger stage 2: air outlet temperature T51	0,1 °C	-990	+3270	INT
iDV35	106	53	74	30075	330	40331	107/106	85	170	Oil system: oil temperature inlet lubrication T66	0,1 °C	-990	+3270	INT
iDV36	108	54	75	30076	331	40332	109/108	86	172	Motor temperature	0,1 °C	-1250	2670	INT
iDV37	110	55	76	30077	332	40333	111/110	87	174	Stage 2: water outlet temperature T13	0,1 °C	-990	+3270	INT
iDV38	112	56	77	30078	333	40334	113/112	88	176	Primary water system: water outlet temperature T14	0,1 °C	-990	+3270	INT
iDV39	114	57	78	30079	334	40335	115/114	89	178	Current cut-out pressure pA (pB)	mbar			INT
iDV40	116	58	79	30080	335	40336	117/116	90	180	Air outlet temperature T100	0,1 °C	-990	+3270	INT
iDV41	118	59	80	30081	336	40337	119/118	91	182	Compressor model / free				INT
iDV42	120	60	81	30082	337	40338	121/120	92	184	Compressor start delay	s			INT
iDV43	122	61	82	30083	338	40339	123/122	93	186	Duty cycle	1%			INT
iDV44	124	62	83	30084	339	40340	125/124	94	188	Current motor speed	1/min			INT
usiSlaveMsgFooter	126	63	84	30085	340	40341	127/126	95	190	Message footer slave				UINT

1. Compressor ==>>> Master Controller

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* since manuf. date MFGDT: 2010/11

JSON-RPC	Sigma Control 2								SAM		Description	Example		
	Profibus		Modbus **				DeviceNet EtherNet/IP	Send/Receive		HEX		Value	Result	
	BDSnd	S7 DBB	S5 DW	Input address	Input Register	Protocol address	Holding Register	Byte	Register MSB/LSB					S5 DW
%	0	DL0	%	%	%	%		%	DL96	192	Profibus: System status 1			
%	1	DR0	%	%	%	%		%	DR96	193	Profibus: System status 2			
%	2	DL1	%	%	%	%		%	DL97	194	Profibus: System status 3			
%	3	DR1	%	%	%	%		%	DR97	195	Profibus: DP master address	01	1	
%	4	DL2	%	%	%	%		%	DL98	196	Profibus: Manufacturer code: PNO number	0C		Kaeser PNO Nr.:
%	5	DR2	%	%	%	%		%	DR98	197	Profibus: Manufacturer code: PNO number	EC		0CEC
%	6	DL3	%	%	%	%		%	DL99	198	Profibus: Header-Byte specific diagnostic data			
%	7	DR3	1	30002	257	40258	MSB	%	DR99	199	SC2 MCS Kaeser SN: 5th digit, 6th digit *	62	62	SC2 MCS
%	8	DL4	1	30002	257	40258	LSB	%	DL100	200	SC2 MCS Kaeser SN: 3rd digit, 4th digit *	30	30	Kaeser SN: 003062
%	9	DR4	2	30003	258	40259	MSB	%	DR100	201	SC2 MCS Kaeser SN: 1st digit, 2nd digit *	00	00	
%	10	DL5	2	30003	258	40259	LSB	%	DL101	202	reserved	00		
%	11	DR5	8	30009	264	40265	MSB	%	DR101	203	SC2 MCS Kaeser PN: space (ASCII)	20	SP	SC2 MCS Kaeser PN: 7.7601P0
%	12	DL6	8	30009	264	40265	LSB	%	DL102	204	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	13	DR6	9	30010	265	40266	MSB	%	DR102	205	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	14	DL7	9	30010	265	40266	LSB	%	DL103	206	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	15	DR7	10	30011	266	40267	MSB	%	DR103	207	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	16	DL8	10	30011	266	40267	LSB	%	DL104	208	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	17	DR8	11	30012	267	40268	MSB	%	DR104	209	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	18	DL9	11	30012	267	40268	LSB	%	DL105	210	SC2 MCS Kaeser PN: space (ASCII)	20	SP	
%	19	DR9	4	30005	260	40261	MSB	%	DR105	211	SC2 MCS Kaeser PN: 8th digit (ASCII)	30	0	
%	20	DL10	4	30005	260	40261	LSB	%	DL106	212	SC2 MCS Kaeser PN: 7th digit (ASCII)	50	P	
%	21	DR10	5	30006	261	40262	MSB	%	DR106	213	SC2 MCS Kaeser PN: 6th digit (ASCII)	31	1	
%	22	DL11	5	30006	261	40262	LSB	%	DL107	214	SC2 MCS Kaeser PN: 5th digit (ASCII)	30	0	
%	23	DR11	6	30007	262	40263	MSB	%	DR107	215	SC2 MCS Kaeser PN: 4th digit (ASCII)	36	6	
%	24	DL12	6	30007	262	40263	LSB	%	DL108	216	SC2 MCS Kaeser PN: 3rd digit (ASCII)	37	7	
%	25	DR12	7	30008	263	40264	MSB	%	DR108	217	SC2 MCS Kaeser PN: 2nd digit (ASCII)	2E	.	
%	26	DL13	7	30008	263	40264	LSB	%	DL109	218	SC2 MCS Kaeser PN: 1st digit (ASCII)	37	7	

1. Compressor ==>>> Master Controller

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* since manuf. date MFGDT: 2010/11

JSON-RPC	Sigma Control 2								SAM		Description	Example		
	Profibus		Modbus **					DeviceNet EtherNet/IP	Send/Receive			HEX	Value	Result
	BDSnd	S7 DBB	S5 DW	Input		Holding			Register MSB/LSB	S5 DW				
Protocol address		Input Register	Protocol address	Holding Register	Byte									
%	27	DR13	12	30013	268	40269	MSB	%	DR109	219	reserved	00		
%	28	DL14	12	30013	268	40269	LSB	%	DL110	220	reserved	00		
%	29	DR14	13	30014	269	40270	MSB	%	DR110	221	reserved	00		
%	30	DL15	13	30014	269	40270	LSB	%	DL111	222	reserved	00		
%	%	%	14	30015	270	40271	MSB	%	%	%	reserved	00		
%	%	%	14	30015	270	40271	LSB	%	%	%	reserved	00		
%	%	%	15	30016	271	40272	MSB	%	%	%	reserved	00		
%	%	%	15	30016	271	40272	LSB	%	%	%	reserved	00		
%	%	%	16	30017	272	40273	MSB	%	%	%	reserved	00		
%	%	%	16	30017	272	40273	LSB	%	%	%	reserved	00		
%	%	%	17	30018	273	40274	MSB	%	%	%	reserved	00		
%	%	%	17	30018	273	40274	LSB	%	%	%	reserved	00		
%	%	%	18	30019	274	40275	MSB	%	%	%	Modbus: Controller identifier (ASCII)	20	SP	SC2
%	%	%	18	30019	274	40275	LSB	%	%	%	Modbus: Controller identifier (ASCII)	53	S	
%	%	%	19	30020	275	40276	MSB	%	%	%	Modbus: Controller identifier (ASCII)	43	C	
%	%	%	19	30020	275	40276	LSB	%	%	%	Modbus: Controller identifier (ASCII)	32	2	

SIGMA CONTROL 2 Process Map

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Table 1.1

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC BDSnd	Profibus/-net		Sigma Control 2 Modbus**				Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX			
xAM1	1.0	0	21	30022	277	40278	0	0	0	32	65.0	0001	dp4 < ... bar start	The compressor drive motor is turning in the wrong direction.
xAM2	1.1	0	21	30022	277	40278	1	0	1	32	65.1	0002	Compressor motor T > ... °C	Compressor drive motor overheated. (PT100)
xAM3	1.2	0	21	30022	277	40278	2	0	2	32	65.2	0003	Test p5: p5 > pRV	Test of pressure relief valve: pressure relief valve does not open
xAM4	1.3	0	21	30022	277	40278	3	0	3	32	65.3	0004	EMERGENCY STOP	EMERGENCY STOP control device actuated.
xAM5	1.4	0	21	30022	277	40278	4	0	4	32	65.4	0005	Diagnostics group alarm	A diagnostic alarm occurred.
xAM6	1.5	0	21	30022	277	40278	5	0	5	32	65.5	0006	T31 < ... °C run	Coolant temperature too low.
xAM7	1.6	0	21	30022	277	40278	6	0	6	32	65.6	0007	Mains monitor	Power supply fault (separate network monitoring module)
xAM8	1.7	0	21	30022	277	40278	7	0	7	32	65.7	*	Alarm during motor standstill	Group alarm "..... Stop" or "..... h Stop"
xAM9	0.0	0	21	30022	277	40278	8	1	0	32	64.0	0009	Sigma Control T ‡	Permissible internal temperature for SIGMA CONTROL 2 MCS or IOM exceeded.
xAM10	0.1	0	21	30022	277	40278	9	1	1	32	64.1	0010	pRV-p5 < ... bar always	Network actual pressure p5 too high (Blow-off protection).
xAM11	0.2	0	21	30022	277	40278	10	1	2	32	64.2	*	Main fan overcurrent	Overload shut-down of the main fan motor.
xAM12	0.3	0	21	30022	277	40278	11	1	3	32	64.3	0012	T31-T10 < ... °C run	Measuring error. May be the temperature sensor is not in correct position.
xAM13	0.4	0	21	30022	277	40278	12	1	4	32	64.4	0013	Compressor motor overload	Overload shut-down of the compressor drive motor.
xAM14	0.5	0	21	30022	277	40278	13	1	5	32	64.5	0014	Auxiliary fan overcurrent	Overload shut-down of the auxilliary fan motor.
xAM15	0.6	0	21	30022	277	40278	14	1	6	32	64.6	0015	p4a/p3a > ... load	Check valve is defective. Final pressure too high.
xAM16	0.7	0	21	30022	277	40278	15	1	7	32	64.7	0016	T11-T10 < ... °C load	Coolant flow rate in the intercooler too high.

SIGMA CONTROL 2 Process Map

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Table 1.1

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	Modbus S5 Bit	Register	Bit	S5 DW	S7 DBX			
xAM17		3.0	1	22	30023	278	40279	0	2	0	33	67.0	0017	p4 > ... bar idle	Pressure outlet stage 2 too high in idle mode / Blow-off silencer clogged. Venting valve defective.
xAM18		3.1	1	22	30023	278	40279	1	2	1	33	67.1	0018	p4 < ... bar load	Pressure outlet stage 2 too low in load mode/ Empty compr. air network. Compr. motor is turning in the wrong direction or is stopped.
xAM19		3.2	1	22	30023	278	40279	2	2	2	33	67.2	0019	p4 > ... bar always	Pressure outlet stage 2 too high / System pressure too high. Pressure differential across the air treatment components too high.
xAM20		3.3	1	22	30023	278	40279	3	2	3	33	67.3	0020	p66 > ... bar always	Oil pressure lubrication too high / Pressure reducer defective.
xAM21		3.4	1	22	30023	278	40279	4	2	4	33	67.4	0021	p66 < ... bar run	Oil pressure lubrication too low / Oil level too low. Pressure reducer defective. Leaks in the machine.
xAM22		3.5	1	22	30023	278	40279	5	2	5	33	67.5	0022	p63 > ... bar run	Oil pressure control too high / Oil too viscous or too cold. Overflow valve defective or incorrectly set.
xAM23		3.6	1	22	30023	278	40279	6	2	6	33	67.6		free	
xAM24		3.7	1	22	30023	278	40279	7	2	7	33	67.7	0024	T13 > ... °C always	Cooling water deficiency
xAM25		2.0	1	22	30023	278	40279	8	3	0	33	66.0	0025	p14 > ... bar always	Primary water system pressure too high.
xAM26		2.1	1	22	30023	278	40279	9	3	1	33	66.1	0026	p14 < ... bar run	Primary water system pressure too low.
xAM27		2.2	1	22	30023	278	40279	10	3	2	33	66.2	0027	p3a < ... bar always	Inlet valve incorrectly set. SFC: Switching to IDLE at high drive motor speed.
xAM28		2.3	1	22	30023	278	40279	11	3	3	33	66.3	0028	p3 > ... bar idle	Inlet valve incorrectly set.
xAM29		2.4	1	22	30023	278	40279	12	3	4	33	66.4	0029	p3a/p1a ... load	Inlet valve is not opening. Leakage between the compression stages. Machine does not start. Air intake restricted.
xAM30		2.5	1	22	30023	278	40279	13	3	5	33	66.5	0030	p3 > ... bar always	Inlet valve incorrectly set.
xAM31		2.6	1	22	30023	278	40279	14	3	6	33	66.6	*	Rotary dryer Group alarm	Fault in rotary dryer
xAM32		2.7	1	22	30023	278	40279	15	3	7	33	66.7	0032	T12T11M-T14 > ... °C run	No cooling. Incorrect coolant flow direction. Primary water system control valve V14 overcompensating. T14w set too high.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.1

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xAM33		5.0	2	23	30024	279	40280	0	4	0	34	69.0	0033	Intercooler condensate drain	The condensate drain on the compressed air intercooler is defective.
xAM34		5.1	2	23	30024	279	40280	1	4	1	34	69.1	0034 0099	Compressor motor Mains contactor not on	Mains contactor for compressor motor does not close.
xAM35		5.2	2	23	30024	279	40280	2	4	2	34	69.2	0035	Cabinet fan I ‡	Overload shut-down of the control cabinet fan motor.
xAM36		5.3	2	23	30024	279	40280	3	4	3	34	69.3	*	WS valve V10 fault	Control valve V10 in manual mode or incorrectly set or defective.
xAM37		5.4	2	23	30024	279	40280	4	4	4	34	69.4	0037	T12-T10 < ... °C load	Coolant flow rate in the aftercooler too high. Hot air connection operating.
xAM38		5.5	2	23	30024	279	40280	5	4	5	34	69.5	0038	T12 > ... °C always	Problems in the primary water system. Set point value T14w set too high. Control valve V14 defective.
xAM39		5.6	2	23	30024	279	40280	6	4	6	34	69.6	0039	free	
xAM40		5.7	2	23	30024	279	40280	7	4	7	34	69.7	0040 0100	Compressor motor Mains contactor not off	Mains contactor for compressor motor does not open.
xAM41		4.0	2	23	30024	279	40280	8	5	0	34	68.0	0041	Mains voltage ‡	Second power failure.
xAM42		4.1	2	23	30024	279	40280	9	5	1	34	68.1	*	WS valve V12 fault	Control valve V12 in manual mode or incorrectly set or defective.
xAM43		4.2	2	23	30024	279	40280	10	5	2	34	68.2	0043	p4a/p3a > ... always	Valves defective or incorrectly set. Blow-off silencer defective.
xAM44		4.3	2	23	30024	279	40280	11	5	3	34	68.3	0044	p4-p5 > ... bar always	Check valve is defective.
xAM45		4.4	2	23	30024	279	40280	12	5	4	34	68.4	0045	p4-p5 < ... bar load	Venting valve is defective.
xAM46		4.5	2	23	30024	279	40280	13	5	5	34	68.5	*	Compressor motor FC Group alarm	Fault in frequency converter or soft starter for compressor motor
xAM47		4.6	2	23	30024	279	40280	14	5	6	34	68.6	*	USS bus communication error	RS485-FC: USS bus communication error
xAM48		4.7	2	23	30024	279	40280	15	5	7	34	68.7	*	External switch cell	Fault in the external switch cell, e.g. high voltage cell.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.1

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xAM49		7.0	3	24	30025	280	40281	0	6	0	35	71.0	0049	T1 > ... °C run	Intake air temperature too high.
xAM50		7.1	3	24	30025	280	40281	1	6	1	35	71.1	0050	T2-T1 < ... °C run	Measuring error.
xAM51		7.2	3	24	30025	280	40281	2	6	2	35	71.2	0051	T2 > ... °C always	Intake air temperature too high. Measuring error.
xAM52		7.3	3	24	30025	280	40281	3	6	3	35	71.3	*	Water pump	Water pump: Overcurrent or motortemperature too high
xAM53		7.4	3	24	30025	280	40281	4	6	4	35	71.4	0053	T2T1M > ... °C always	Intake air temperature too high.
xAM54		7.5	3	24	30025	280	40281	5	6	5	35	71.5	0054	T31 > .. °C run	Cooler clogged. Coolant temperature too high. Set point value T10w too high.
xAM55		7.6	3	24	30025	280	40281	6	6	6	35	71.6	0055	T32 > .. °C run	Fault at valve V32
xAM56		7.7	3	24	30025	280	40281	7	6	7	35	71.7	0056	T4 > .. °C always	Coolant temperature too high. Set point value T10w too high.
xAM57		6.0	3	24	30025	280	40281	8	7	0	35	70.0	0057	Model?	Model type ambiguous
xAM58		6.1	3	24	30025	280	40281	9	7	1	35	70.1	0058	Aftercooler condensate drain	The condensate drain on the compressed air aftercooler is defective.
xAM59		6.2	3	24	30025	280	40281	10	7	2	35	70.2	0059	T51 < ... °C load	Coolant temperature too low.
xAM60		6.3	3	24	30025	280	40281	11	7	3	35	70.3	*	WS valve V14 fault	Control valve V14 in manual mode or incorrectly set or defective.
xAM61		6.4	3	24	30025	280	40281	12	7	4	35	70.4	0061	T66 > ... °C run	Oil cooler clogged. Flow rate through the oil cooler too low.
xAM62		6.5	3	24	30025	280	40281	13	7	5	35	70.5	*	Current ‡ or ‡	Free definable analogue current value passed a set limit (Menue <Analog values> <AnMod_I_x>)
xAM63		6.6	3	24	30025	280	40281	14	7	6	35	70.6	0063	T10 > ... °C run	Problems in the cooling water system. Set point value T10w set too high. Control valve V14 defective.
xAM64		6.7	3	24	30025	280	40281	15	7	7	35	70.7	0064	T10 < .. °C run	Temperature of the coolant in the primary water system too low.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.1

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xAM65		9.0	4	25	30026	281	40282	0	8	0	36	73.0	0065	T11 > ... °C always	Problems in the cooling water system. Set point value T14w set too high. Control valve V14 defective.
xAM66		9.1	4	25	30026	281	40282	1	8	1	36	73.1	0066	T4-T32 < ... °C run	Final pressure too low.
xAM67		9.2	4	25	30026	281	40282	2	8	2	36	73.2	0067	T4T32M > ... °C always	Intermediate stage temperature too high. Set point value T10w and/or set point value T14w too high.
xAM68		9.3	4	25	30026	281	40282	3	8	3	36	73.3	0068	T4T32p4 < ... °C load	Measuring error.
xAM69		9.4	4	25	30026	281	40282	4	8	4	36	73.4	0069	T4T32p4 > ... °C load	Measuring error.
xAM70		9.5	4	25	30026	281	40282	5	8	5	36	73.5	0070	T51 > ... °C always	Cooler clogged. Coolant temperature too high. Coolant flow rate too low.
xAM71		9.6	4	25	30026	281	40282	6	8	6	36	73.6	0071	Compressor motor Redundancy contactor on?	Redundancy contactor for compressor motor does not close.
xAM72		9.7	4	25	30026	281	40282	7	8	7	36	73.7	0072	Compressor motor Redundancy contactor off?	Redundancy contactor for compressor motor does not open.
xAM73		8.0	4	25	30026	281	40282	8	9	0	36	72.0	0073	External message 1	External binary message 1
xAM74		8.1	4	25	30026	281	40282	9	9	1	36	72.1	0074	External message 2	External binary message 2
xAM75		8.2	4	25	30026	281	40282	10	9	2	36	72.2	0075	External message 3	External binary message 3
xAM76		8.3	4	25	30026	281	40282	11	9	3	36	72.3	0076	External message 4	External binary message 4
xAM77		8.4	4	25	30026	281	40282	12	9	4	36	72.4	0077	External message 5	External binary message 5
xAM78		8.5	4	25	30026	281	40282	13	9	5	36	72.5	0078	External message 6	External binary message 6
xAM79		8.6	4	25	30026	281	40282	14	9	6	36	72.6	*	Pressure ‡ or ‡	Free definable analogue pressure value passed a set limit (Menue <Analog values> <AnMod_p_x>)
xAM80		8.7	4	25	30026	281	40282	15	9	7	36	72.7	*	Temperature ‡ or ‡	Free definable analogue temperature value passed a set limit (Menue <Analog values> <AnMod_T_x>)

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xWM1		11.0	5	26	30027	282	40283	0	10	0	37	75.0		free	
xWM2		11.1	5	26	30027	282	40283	1	10	1	37	75.1	0002	Compressor motor T > ... °C	Drive motor overheating. (PT100)
xWM3		11.2	5	26	30027	282	40283	2	10	2	37	75.2	*	Warning during motor standstill	Group alarm "..... Stop" or "..... h Stop"
xWM4		11.3	5	26	30027	282	40283	3	10	3	37	75.3	0004	p4a/p3a > ... always	Valve defective. Blow-off silencer defective.
xWM5		11.4	5	26	30027	282	40283	4	10	4	37	75.4	*	Rotary dryer Group warning	Warning at rotary dryer occurred.
xWM6		11.5	5	26	30027	282	40283	5	10	5	37	75.5	0006	p4-p5 > ... bar always	Check valve is defective.
xWM7		11.6	5	26	30027	282	40283	6	10	6	37	75.6	0007	Idle period > s	Idle period of compressor motor takes too long.
xWM8		11.7	5	26	30027	282	40283	7	10	7	37	75.7	0008	p4-p5 < ... bar load	Venting valve defective.
xWM9		10.0	5	26	30027	282	40283	8	11	0	37	74.0		free	
xWM10		10.1	5	26	30027	282	40283	9	11	1	37	74.1	0010	p4a/p3a > ... load	Check valve is defective. Final pressure too high.
xWM11		10.2	5	26	30027	282	40283	10	11	2	37	74.2	0011	p64 ↑ motor run	The pressure differential of the oil filter has risen. Oil filter clogged.
xWM12		10.3	5	26	30027	282	40283	11	11	3	37	74.3	0012	T31-T10 > ... °C run	Inadequate cooling (Cooler clogged, lack of cooling water)
xWM13		10.4	5	26	30027	282	40283	12	11	4	37	74.4	0013	p1 < ... bar run	Initial warning: Air filter clogged.
xWM14		10.5	5	26	30027	282	40283	13	11	5	37	74.5	0014	T11 > ... °C always	Coolant temperature in the intermediate cooler high. Temperature in the primary water system changing too quickly.
xWM15		10.6	5	26	30027	282	40283	14	11	6	37	74.6	0015	Bus alarm	The bus link from the Com-Modul interface is interrupted.
xWM16		10.7	5	26	30027	282	40283	15	11	7	37	74.7	0016	T13 > ... °C always	Coolant temperature in the airend high.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xWM17		13.0	6	27	30028	283	40284	0	12	0	38	77.0	0017	T66 < ... °C start	Cooling oil temperature too low.
xWM18		13.1	6	27	30028	283	40284	1	12	1	38	77.1	0018	T3 < ... °C start	Airend temperature low.
xWM19		13.2	6	27	30028	283	40284	2	12	2	38	77.2	0019	p3a < ... bar always	Inlet valve incorrectly set. SFC: Switching to IDLE at high rotation speed of the compressor motor.
xWM20		13.3	6	27	30028	283	40284	3	12	3	38	77.3	0020	p3 > ... bar idle	Inlet valve incorrectly set.
xWM21		13.4	6	27	30028	283	40284	4	12	4	38	77.4	0021	p3a/p1a ... load	Inlet valve is not opening. Leakage between the compression stages. Machine does not start. Air intake restricted.
xWM22		13.5	6	27	30028	283	40284	5	12	5	38	77.5	0022	p3 > ... bar always	Intermediate stage pressure too high.
xWM23		13.6	6	27	30028	283	40284	6	12	6	38	77.6	0023	T31-T10 < ... °C run	Cooler clogged.
xWM24		13.7	6	27	30028	283	40284	7	12	7	38	77.7	0024	T12-T10 < ... °C load	Coolant flow rate in the aftercooler high. Hot air connection operating.
xWM25		12.0	6	27	30028	283	40284	8	13	0	38	76.0	0025	T66 > ... °C run	Oil cooler clogged. Low flow rate through the oil cooler.
xWM26		12.1	6	27	30028	283	40284	9	13	1	38	76.1	0026	T66 < ... °C run	Oil level too low. Thermal element defective. Oil pump defective. Pressure regulating valve defective.
xWM27		12.2	6	27	30028	283	40284	10	13	2	38	76.2	0027	p66 > ... bar always	Pressure reducer defective.
xWM28		12.3	6	27	30028	283	40284	11	13	3	38	76.3	0028	p66 < ... bar run	Oil level too low. Pressure reducer defective. Leaks in the machine.
xWM29		12.4	6	27	30028	283	40284	12	13	4	38	76.4	0029	p63 > ... bar always	Oil too viscous. Overflow valve defective.
xWM30		12.5	6	27	30028	283	40284	13	13	5	38	76.5	0030	p63 < ... bar run	Oil too viscous or too cold. Overflow valve defective or incorrectly set.
xWM31		12.6	6	27	30028	283	40284	14	13	6	38	76.6	0031	T10 > ... °C run	Primary water system: Temperature of the coolant too high. Temperature/pressure ratios changing too quickly.
xWM32		12.7	6	27	30028	283	40284	15	13	7	38	76.7	0032	T10 < ... °C run	Temperature of the coolant in the primary water system low.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xWM33		15.0	7	28	30029	284	40285	0	14	0	39	79.0	0033	p4 > ... bar idle	Blow-off silencer clogged. Venting valve defective.
xWM34		15.1	7	28	30029	284	40285	1	14	1	39	79.1	0034	p4 < ... bar load	Pressure outlet stage 2 too low in load mode/ Empty compr. air network. Compr. motor is turning in the wrong direction or is stopped.
xWM35		15.2	7	28	30029	284	40285	2	14	2	39	79.2	0035	p4 > ... bar always	System pressure too high. High pressure drop over the air treatment components.
xWM36		15.3	7	28	30029	284	40285	3	14	3	39	79.3	0036	Motor starts /h	The permissible number of compressor motor starts was exceeded in the last 60 minutes.
xWM37		15.4	7	28	30029	284	40285	4	14	4	39	79.4	0037	Motor starts /d	The permissible number of compressor motor starts was exceeded in the last 24 hours.
xWM38		15.5	7	28	30029	284	40285	5	14	5	39	79.5	0038	T1 > ... °C run	High intake air temperature.
xWM39		15.6	7	28	30029	284	40285	6	14	6	39	79.6	0039	T2-T1 < ... °C run	Measuring error.
xWM40		15.7	7	28	30029	284	40285	7	14	7	39	79.7	0040	T2 > ... °C always	High intake air temperature. Measuring error.
xWM41		14.0	7	28	30029	284	40285	8	15	0	39	78.0	0041	Mains voltage ↓	First power failure: The machine is automatically restarted.
xWM42		14.1	7	28	30029	284	40285	9	15	1	39	78.1	0042	T12 > ... °C always	Coolant temperature in the aftercooler high. Temperature in the primary water system changing too quickly.
xWM43		14.2	7	28	30029	284	40285	10	15	2	39	78.2	0043	p100 > pE run	Increased cut-out pressure exceeded. The external load control has not switched to idle or for SFC: dpFC is set too high.
xWM44		14.3	7	28	30029	284	40285	11	15	3	39	78.3	0044	T12#T11 > ... °C load	Cooling water distribution incorrectly set.
xWM45		14.4	7	28	30029	284	40285	12	15	4	39	78.4		free	
xWM46		14.5	7	28	30029	284	40285	13	15	5	39	78.5	0046	pN < ... bar load	Network pressure has fallen below the set 'low' value. / Air consumption too high.
xWM47		14.6	7	28	30029	284	40285	14	15	6	39	78.6	0047	T2T1M > ... °C always	High intake air temperature.
xWM48		14.7	7	28	30029	284	40285	15	15	7	39	78.7	0048	T51 < ... °C load	Coolant temperature too low.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xWM49		17.0	8	29	30030	285	40286	0	16	0	40	81.0		reserved	
xWM50		17.1	8	29	30030	285	40286	1	16	1	40	81.1	0050	T31 > ... °C run	Cooler clogged. Coolant temperature high. Set point value T10 high.
xWM51		17.2	8	29	30030	285	40286	2	16	2	40	81.2	0051	T31-T1 > ... °C fan	Message during run of main fan / Cooler clogged. Coolant temperature high.
xWM52		17.3	8	29	30030	285	40286	3	16	3	40	81.3	0052	T11-T10 < ... °C load	Coolant flow rate in the intercooler high.
xWM53		17.4	8	29	30030	285	40286	4	16	4	40	81.4	0053	p14 < ... bar always	Primary water system: pressure too low or system leaking.
xWM54		17.5	8	29	30030	285	40286	5	16	5	40	81.5	0054	T4 > ... °C always	Coolant temperature high. Set point value T10w high.
xWM55		17.6	8	29	30030	285	40286	6	16	6	40	81.6	0055	T12T11M-T14 > ... °C run	No cooling. Incorrect coolant flow direction. Primary water system control valve overcompensating.
xWM56		17.7	8	29	30030	285	40286	7	16	7	40	81.7	0056	T4T3M > .. °C always	Intermediate stage temperature high. Set point value T10w and/or set point value T14w high.
xWM57		16.0	8	29	30030	285	40286	8	17	0	40	80.0	0057	T4T3p4 < ... °C load	Measuring error.
xWM58		16.1	8	29	30030	285	40286	9	17	1	40	80.1	0058	SC2 <=> SC2 Communication error	SC2 to SC2 communication is interrupted.
xWM59		16.2	8	29	30030	285	40286	10	17	2	40	80.2	0059	T1 < ... °C start	Low intake air temperature.
xWM60		16.3	8	29	30030	285	40286	11	17	3	40	80.3	0060	T31 < ... °C run	Coolant temperature too low.
xWM61		16.4	8	29	30030	285	40286	12	17	4	40	80.4	0061	T4T3p4 > ... °C load	Measuring error
xWM62		16.5	8	29	30030	285	40286	13	17	5	40	80.5	*	Current ‡ or ‡	Free definable analogue current value passed a set limit (Menue <Analog values> <AnMod_I_x>)
xWM63		16.6	8	29	30030	285	40286	14	17	6	40	80.6	0063	T32 < ... °C start	Coolant temperature too low.
xWM64		16.7	8	29	30030	285	40286	15	17	7	40	80.7	0210 0230	Compressor motor FC group warning	Compressor motor FC Service mode active or fault at AI0 or AI1 of the frequency converter

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xWM65		19.0	9	30	30031	286	40287	0	18	0	41	83.0	0065	T51 > ... °C always	Cooler clogged. Coolant temperature high. Low coolant flow rate.
xWM66		19.1	9	30	30031	286	40287	1	18	1	41	83.1	0066	T100 > .. °C always	Outlet temperature too high
xWM67		19.2	9	30	30031	286	40287	2	18	2	41	83.2	0067	T51-T1 > ... °C fan	Message during run of main fan / Insufficient cooling.
xWM68		19.3	9	30	30031	286	40287	3	18	3	41	83.3	0068	Aftercooler condensate drain	The condensate drain on the compressed air aftercooler is defective.
xWM69		19.4	9	30	30031	286	40287	4	18	4	41	83.4	0069	Intercooler condensate drain	The condensate drain on the compressed air intercooler is defective.
xWM70		19.5	9	30	30031	286	40287	5	18	5	41	83.5	0070	T51-T10 < ... °C load	Insufficient cooling.
xWM71		19.6	9	30	30031	286	40287	6	18	6	41	83.6	0071	T32 > .. °C run	Option H21: Inlet temperature at stage 2 too high
xWM72		19.7	9	30	30031	286	40287	7	18	7	41	83.7	0072	T2-T1 < .. °C always	Measuring error
xWM73		18.0	9	30	30031	286	40287	8	19	0	41	82.0	0073	External message 1	External binary message 1
xWM74		18.1	9	30	30031	286	40287	9	19	1	41	82.1	0074	External message 2	External binary message 2
xWM75		18.2	9	30	30031	286	40287	10	19	2	41	82.2	0075	External message 3	External binary message 3
xWM76		18.3	9	30	30031	286	40287	11	19	3	41	82.3	0076	External message 4	External binary message 4
xWM77		18.4	9	30	30031	286	40287	12	19	4	41	82.4	0077	External message 5	External binary message 5
xWM78		18.5	9	30	30031	286	40287	13	19	5	41	82.5	0078	External message 6	External binary message 6
xWM79		18.6	9	30	30031	286	40287	14	19	6	41	82.6	*	Pressure ‡ or ‡	Free definable analogue pressure value passed a set limit (Menue <Analog values> <AnMod_p_x>)
xWM80		18.7	9	30	30031	286	40287	15	19	7	41	82.7	*	Temperature ‡ or ‡	Free definable analogue temperature value passed a set limit (Menue <Analog values> <AnMod_T_x>)

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.3

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages and maintenance messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

M Maintenance message

O Operational message

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			
xOM1		21.0	10	31	30032	287	40288	0	20	0	42	85.0	M 0456	Valve inspection h ‡	Maintenance interval for the next valve inspection has expired.
xOM2		21.1	10	31	30032	287	40288	1	20	1	42	85.1	M 0455	Main fan Bearing change h ‡	Maintenance interval for re-lubricating the main fan motor bearings has expired.
xOM3		21.2	10	31	30032	287	40288	2	20	2	42	85.2	M 460	Main fan Bearing lube h ‡	Maintenance interval for greasing the main fan motor bearings has expired.
xOM4		21.3	10	31	30032	287	40288	3	20	3	42	85.3	M 461	Fan T100 Bearing change h ‡	Maintenance interval for the T100 fan's motor bearing has expired.
xOM5		21.4	10	31	30032	287	40288	4	20	4	42	85.4	M 470	RD fan motor Bearing lubrication h ‡	Maintenance interval for greasing the rotary dryer fan motor bearings has expired.
xOM6		21.5	10	31	30032	287	40288	5	20	5	42	85.5	M 471	RD fan motor Bearing change h ‡	Maintenance interval for the rotary dryer fan's motor bearing has expired.
xOM7		21.6	10	31	30032	287	40288	6	20	6	42	85.6	M	frei	
xOM8		21.7	10	31	30032	287	40288	7	20	7	42	85.7	M 472	RD drum motor oil change h ‡	Maintenance interval for the drum motor oil has expired.
xOM9		20.0	10	31	30032	287	40288	8	21	0	42	84.0	M 459	Coupling change h ‡	Maintenance interval for coupling has expired.
xOM10		20.1	10	31	30032	287	40288	9	21	1	42	84.1	M 0454	Oil change h ‡	The oil life has expired.
xOM11		20.2	10	31	30032	287	40288	10	21	2	42	84.2	M 0453	Oil filter h ‡	The oil filter life has expired.
xOM12		20.3	10	31	30032	287	40288	11	21	3	42	84.3	M 0450	Air filter h ‡	The air filter life has expired.
xOM13		20.4	10	31	30032	287	40288	12	21	4	42	84.4	M 0457	Compressor motor Bearing lube h ‡	The maintenance interval for re-greasing the compressor drive motor bearings has expired.
xOM14		20.5	10	31	30032	287	40288	13	21	5	42	84.5	M 0451	Coupling inspection h ‡	The interval for the next coupling inspection has expired.
xOM15		20.6	10	31	30032	287	40288	14	21	6	42	84.6	M 0452	Compressor motor Bearing change h ‡	Operational life of the compressor drive motor bearings has expired.
xOM16		20.7	10	31	30032	287	40288	15	21	7	42	84.7	M 0458	Electrical equipment h ‡	The maintenance interval for checking electrical equipment and wiring has expired.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.3

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages and maintenance messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

M Maintenance message

O Operational message

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2						DeviceNet EtherNet/IP		SAM		Message		Description / possible cause
			Modbus**				Send/Receive								
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address	Holding Register	Modbus S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)	
xOM17	23.0	11	32	30033	288	40289	0	22	0	43	87.0	O 0017	External message 1	External binary message 1	
xOM18	23.1	11	32	30033	288	40289	1	22	1	43	87.1	O 0018	External message 2	External binary message 2	
xOM19	23.2	11	32	30033	288	40289	2	22	2	43	87.2	O 0019	External message 3	External binary message 3	
xOM20	23.3	11	32	30033	288	40289	3	22	3	43	87.3	O 0020	External message 4	External binary message 4	
xOM21	23.4	11	32	30033	288	40289	4	22	4	43	87.4	O 0021	External message 5	External binary message 5	
xOM22	23.5	11	32	30033	288	40289	5	22	5	43	87.5	O 0022	External message 6	External binary message 6	
xOM23	23.6	11	32	30033	288	40289	6	22	6	43	87.6	O *	Pressure ‡ or ‡	Free definable analogue pressure value passed a set limit (Menue <Analog values> <AnMod_p_x>)	
xOM24	23.7	11	32	30033	288	40289	7	22	7	43	87.7	O *	Temperature ‡ or ‡	Free definable analogue temperature value passed a set limit (Menue <Analog values> <AnMod_T_x>)	
xOM25	22.0	11	32	30033	288	40289	8	23	0	43	86.0	O 0025	Test DO active	The DO test is active	
xOM26	22.1	11	32	30033	288	40289	9	23	1	43	86.1	O 0026	p4 > .. bar start	The compressor starts after depressurising.	
xOM27	22.2	11	32	30033	288	40289	10	23	2	43	86.2	O 0027	Power OFF -> ON	It is necessary to switch power supply of SC2 off and on.	
xOM28	22.3	11	32	30033	288	40289	11	23	3	43	86.3	O 0028	DYNAMIC compressor motor T > ... °C	Control mode DYNAMIC: The temperature of the compressor motor is too high.	
xOM29	22.4	11	32	30033	288	40289	12	23	4	43	86.4	O *	Current ‡ or ‡	Free definable analogue current value passed a set limit (Menue <Analog values> <AnMod_I_x>)	
xOM30	22.5	11	32	30033	288	40289	13	23	5	43	86.5	O 0030	Voltage restored	SC2 has been feed with supply voltage.	
xOM31	22.6	11	32	30033	288	40289	14	23	6	43	86.6	O 0031	Control mode DYNAMIC -> QUADRO	Control mode was automatically switched to QUADRO	
xOM32	22.7	11	32	30033	288	40289	15	23	7	43	86.7	M	free		

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.3

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages and maintenance messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

M Maintenance message

O Operational message

* Summary of SC2 messages

JSON-RPC BDSnd	Profibus/-net		Sigma Control 2 Modbus**						DeviceNet EtherNet/IP		SAM Send/Receive		Message		Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	Modbus S5 Bit	Register	Bit	S5 DW	S7 DBX	ID			
xOM33	25.0	12	33	30034	289	40290	0	24	0	44	89.0	O 0033	Machine report		Life sign has been sent.
xOM34	25.1	12	33	30034	289	40290	1	24	1	44	89.1	O 0034	E-mail send unsuccessful!		E-Mail cannot be sent.
xOM35	25.2	12	33	30034	289	40290	2	24	2	44	89.2	O 0035	Water pump runs		Water pump is running
xOM36	25.3	12	33	30034	289	40290	3	24	3	44	89.3	M	free		
xOM37	25.4	12	33	30034	289	40290	4	24	4	44	89.4	M	free		
xOM38	25.5	12	33	30034	289	40290	5	24	5	44	89.5	M	free		
xOM39	25.6	12	33	30034	289	40290	6	24	6	44	89.6	M	free		
xOM40	25.7	12	33	30034	289	40290	7	24	7	44	89.7	O 0040	free		
xOM41	24.0	12	33	30034	289	40290	8	24	0	44	88.0	O 0041	free		
xOM42	24.1	12	33	30034	289	40290	9	24	1	44	88.1	O 0042	free		
xOM43	24.2	12	33	30034	289	40290	10	24	2	44	88.2	O 0043	free		
xOM44	24.3	12	33	30034	289	40290	11	24	3	44	88.3	O 0044	free		
xOM45	24.4	12	33	30034	289	40290	12	24	4	44	88.4	O 0045	free		
xOM46	24.5	12	33	30034	289	40290	13	24	5	44	88.5	O 0046	free		
xOM47	24.6	12	33	30034	289	40290	14	24	6	44	88.6	O 0047	free		
xOM48	24.7	12	33	30034	289	40290	15	24	7	44	88.7	M 0469	Annual maintenance		Appears one year after last maintenance.

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.4

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

Sigma Control 2													Description		Signal	
JSON-RPC	Profibus/-net		Modbus**				Modbus EtherNet/IP	SAM								
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	DeviceNet Register	Bit	S5 DW				
xBS0	27.0	13	34	30035	290	40291	0	26	0	45	91.0	Controller on (LED at key "I" on) + Remote on + Bus (Receive)	yes	no		
xBS1	27.1	13	34	30035	290	40291	1	26	1	45	91.1	Motor running	yes	no		
xBS2	27.2	13	34	30035	290	40291	2	26	2	45	91.2	Load run	yes	no		
xBS3	27.3	13	34	30035	290	40291	3	26	3	45	91.3	Idle	yes	no		
xBS4	27.4	13	34	30035	290	40291	4	26	4	45	91.4	Local mode (LED at key "Remote" off)	yes	no		
xBS5	27.5	13	34	30035	290	40291	5	26	5	45	91.5	Controller on (LED at key "I" on)	yes	no		
xBS6	27.6	13	34	30035	290	40291	6	26	6	45	91.6	Group alarm	yes	no		
xBS7	27.7	13	34	30035	290	40291	7	26	7	45	91.7	Group warning	yes	no		
xBS8	26.0	13	34	30035	290	40291	8	27	0	45	90.0	Compressor ready for load	yes	no		
xBS9	26.1	13	34	30035	290	40291	9	27	1	45	90.1	Compressor ON (Motor is running or ready for operation)	yes	no		
xBS10	26.2	13	34	30035	290	40291	10	27	2	45	90.2	Automatic restart after power on	activated	deactivated		
xBS11	26.3	13	34	30035	290	40291	11	27	3	45	90.3	Message acknowledgement by bus master	activated	deactivated		
xBS12	26.4	13	34	30035	290	40291	12	27	4	45	90.4	Main fan running	running	off		
xBS13	26.5	13	34	30035	290	40291	13	27	5	45	90.5	Auxiliary fan running	running	off		
xBS14	26.6	13	34	30035	290	40291	14	27	6	45	90.6	Maintenance message is active (Collective message xOM1 ... xOM16, 32, 36...39, 48)	yes	no		
xBS15	26.7	13	34	30035	290	40291	15	27	7	45	90.7					

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.4

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2								SAM		Description	Signal		
	Profibus/-net		Modbus**				DeviceNet EtherNet/IP		Send/Receive					
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address	Holding Register	S5 Bit	Register	Bit		S5 DW	S7 DBX	"1"
xBS16	29.0	14	35	30036	291	40292	0	28	0	46	93.0	Mode <i>compressor ON from "I" key only</i> is active	yes	no
xBS17	29.1	14	35	30036	291	40292	1	28	1	46	93.1	Mode <i>compressor ON from "I" key and clock</i> is active	yes	no
xBS18	29.2	14	35	30036	291	40292	2	28	2	46	93.2			
xBS19	29.3	14	35	30036	291	40292	3	28	3	46	93.3			
xBS20	29.4	14	35	30036	291	40292	4	28	4	46	93.4	Mode <i>compressor ON from "I" key and external contact</i> is active	yes	no
xBS21	29.5	14	35	30036	291	40292	5	28	5	46	93.5	Mode <i>compressor ON from "I" key and clock or external contact</i> is active	yes	no
xBS22	29.6	14	35	30036	291	40292	6	28	6	46	93.6	Mode <i>compressor ON from "I" key and bus master</i> is active	yes	no
xBS23	29.7	14	35	30036	291	40292	7	28	7	46	93.7	Mode compressor ON (xBS16..xBS22) set from	bus master	local
xBS24	28.0	14	35	30036	291	40292	8	29	0	46	92.0	DUAL control mode is active	yes	no
xBS25	28.1	14	35	30036	291	40292	9	29	1	46	92.1	QUADRO control mode is active	yes	no
xBS26	28.2	14	35	30036	291	40292	10	29	2	46	92.2			
xBS27	28.3	14	35	30036	291	40292	11	29	3	46	92.3	DYNAMIC control mode is active	yes	no
xBS28	28.4	14	35	30036	291	40292	12	29	4	46	92.4			
xBS29	28.5	14	35	30036	291	40292	13	29	5	46	92.5			
xBS30	28.6	14	35	30036	291	40292	14	29	6	46	92.6			
xBS31	28.7	14	35	30036	291	40292	15	29	7	46	92.7	Control mode (xBS24 - xBS28) set from	bus master	local

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.4

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2								SAM		Description	Signal		
	Profibus/-net		Modbus**				DeviceNet EtherNet/IP		Send/Receive					
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address	Holding Register	S5 Bit	Register	Bit		S5 DW	S7 DBX	"1"
xBS32	31.0	15	36	30037	292	40293	0	30	0	47	95.0			
xBS33	31.1	15	36	30037	292	40293	1	30	1	47	95.1			
xBS34	31.2	15	36	30037	292	40293	2	30	2	47	95.2			
xBS35	31.3	15	36	30037	292	40293	3	30	3	47	95.3			
xBS36	31.4	15	36	30037	292	40293	4	30	4	47	95.4	Mode <i>load-idle signal from external contact</i> is active	yes	no
xBS37	31.5	15	36	30037	292	40293	5	30	5	47	95.5			
xBS38	31.6	15	36	30037	292	40293	6	30	6	47	95.6	Mode <i>load-idle signal from bus master</i> is active	yes	no
xBS39	31.7	15	36	30037	292	40293	7	30	7	47	95.7	Mode <i>load-idle signal (xBS32..xBS47) set from</i>	bus master	local
xBS40	30.0	15	36	30037	292	40293	8	31	0	47	94.0	Mode <i>load-idle signal from pressure regualtor, pA permanent</i> is active	yes	no
xBS41	30.1	15	36	30037	292	40293	9	31	1	47	94.1	Mode <i>load-idle signal from pressure regualtor, pB permanent</i> is active	yes	no
xBS42	30.2	15	36	30037	292	40293	10	31	2	47	94.2	Mode <i>load-idle signal from pressure regualtor, pA/pB change via clock</i> is active	yes	no
xBS43	30.3	15	36	30037	292	40293	11	31	3	47	94.3	Mode <i>load-idle signal from pressure regualtor, pA/pB change via timer</i> is active	yes	no
xBS44	30.4	15	36	30037	292	40293	12	31	4	47	94.4			
xBS45	30.5	15	36	30037	292	40293	13	31	5	47	94.5	Mode <i>load-idle signal from pressure regualtor, pA/pB change via external contact</i> is active	yes	no
xBS46	30.6	15	36	30037	292	40293	14	31	6	47	94.6	Mode <i>load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet)</i> is active	yes	no
xBS47	30.7	15	36	30037	292	40293	15	31	7	47	94.7			

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 1.4

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC	Sigma Control 2										Description	Signal		
	Profibus/-net		Modbus**				DeviceNet EtherNet/IP	Send/Receive						
	BDSnd	S7 DBX	S5 DW	Input address	Input Register	Holding address		Holding Register	S5 Bit	Register		Bit	S5 DW	S7 DBX
xBS48	33.0	16	37	30038	293	40294	0	32	0	48	97.0	Water system valve V10 opens	opens	
xBS49	33.1	16	37	30038	293	40294	1	32	1	48	97.1	Water system valve V10 closes	closes	
xBS50	33.2	16	37	30038	293	40294	2	32	2	48	97.2	Water system valve V10 open	open	
xBS51	33.3	16	37	30038	293	40294	3	32	3	48	97.3	Water system valve V10 closed	closed	
xBS52	33.4	16	37	30038	293	40294	4	32	4	48	97.4	Water system valve V12 opens	opens	
xBS53	33.5	16	37	30038	293	40294	5	32	5	48	97.5	Water system valve V12 closes	closes	
xBS54	33.6	16	37	30038	293	40294	6	32	6	48	97.6	Water system valve V12 open	open	
xBS55	33.7	16	37	30038	293	40294	7	32	7	48	97.7	Water system valve V10 closed	closed	
xBS56	32.0	16	37	30038	293	40294	8	33	0	48	96.0	Water system valve V14 opens	opens	
xBS57	32.1	16	37	30038	293	40294	9	33	1	48	96.1	Water system valve V14 closes	closes	
xBS58	32.2	16	37	30038	293	40294	10	33	2	48	96.2	Water system valve V14 open	open	
xBS59	32.3	16	37	30038	293	40294	11	33	3	48	96.3	Water system valve V14 mimimum opening	open Minimum	
xBS60	32.4	16	37	30038	293	40294	12	33	4	48	96.4	Water system valve V14 closed	closed	
xBS61	32.5	16	37	30038	293	40294	13	33	5	48	96.5	Water system water pump running	running	off
xBS62	32.6	16	37	30038	293	40294	14	33	6	48	96.6			
xBS63	32.7	16	37	30038	293	40294	15	33	7	48	96.7			

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 2 Process

2. Master Controller ==>>>> Compressor

1) write both values at a time

2) write both values at a time

Table 2: Process data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* negative value

JSON-RPC	Sigma Control 2					SAM		Description	Unit	Format	
	Profibus/Profinet		Modbus**		DeviceNet EtherNet/IP	Send/Receive					
BDRcv	S7 DBW	S5 DW	Protocol address	Holding Register	Register MSB/LSB	S5 DW	S7 DBW				
	0	0	0	40001	1/0	0	0	Receive bit xBS15 - xBS0	Table 2.1	binary	BOOL
	2	1	1	40002	3/2	1	2	Receive bit xBS31 - xBS16	Table 2.1	binary	BOOL
	4	2	2	40003	5/4	2	4	Receive bit xBS47 - xBS32	Table 2.1	binary	BOOL
	6	3	3	40004	7/6	3	6	Receive bit xBS63 - xBS48	Table 2.1	binary	BOOL
iDV0	8	4	4	40005	9/8	4	8	Setpoint pressure pA switching point 1)	if xBS68=1	mbar	INT
iDV1	10	5	5	40006	11/10	5	10	Setpoint pressure pA switching difference * 1)	if xBS68=1	mbar	INT
iDV2	12	6	6	40007	13/12	6	12	Setpoint pressure pB switching point 2)	if xBS70=1	mbar	INT
iDV3	14	7	7	40008	15/14	7	14	Setpoint pressure pB switching difference * 2)	if xBS70=1	mbar	INT
iDV4	16	8	8	40009	17/16	8	16	Set speed	if xBS48=1	n ⁻¹	INT
iDV5	18	9	9	40010	19/18	9	18				
iDV6	20	10	10	40011	21/20	10	20				
iDV7	22	11	11	40012	23/22	11	22				
iDV8	24	12	12	40013	25/24	12	24				
iDV9	26	13	13	40014	27/26	13	26	Actual air receiver system pressure	if xBS77=1	mbar	INT
iDV10	28	14	14	40015	29/28	14	28				
iDV11	30	15	15	40016	31/30	15	30				
iDV12	32	16	16	40017	33/32	16	32				
	34	17	17	40018	35/34	17	34	Receive bit xBS79 - xBS64	Table 2.1	binary	BOOL
	36	18	18	40019	37/36	18	36	Receive bit xBS95 - xBS80	Table 2.1	binary	BOOL
usiMasterMsgHeader	38	19	19	40020	39/38	19	38	Message header master			UINT
iDV13	40	20	20	40021	41/40	20	40				
iDV14	42	21	21	40022	43/42	21	42	Setpoint temperature control T14	if xBS86=1	0,1°C	INT
iDV15	44	22	22	40023	45/44	22	44				
iDV16	46	23	23	40024	47/46	23	46				
iDV17	48	24	24	40025	49/48	24	48				
iDV18	50	25	25	40026	51/50	25	50				
iDV19	52	26	26	40027	53/52	26	52				
iDV20	54	27	27	40028	55/54	27	54				
iDV21	56	28	28	40029	57/56	28	56				
iDV22	58	29	29	40030	59/58	29	58				
iDV23	60	30	30	40031	61/60	30	60				
usiMasterMsgFooter	62	31	31	40032	63/62	31	62	Message footer master			UINT

SIGMA CONTROL 2 Process Map

≥ dry_3.1.1

Table 2 Buspar

2. Master Controller ==>>>> Compressor

Table 2: Busparameter , Modbus only

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

* not for Modbus TCP

Sigma Control 2			Modbus **			DeviceNet EtherNet/IP		SAM Send/Receive		Description	Example		
JSON-RPC	Profibus/Profinet		Protocol	Holding	Bit	Register	MSB/LSB	S5 DW	S7 DBB		HEX	Value	Result
BDSnd	S7 DBB	S5 DW	address	Register									
%	%	%	512	40513		%	%	%	%	Modbus: Slaveaddress	64	100	Slave: 100
%	%	%	513	40514		%	%	%	%	Modbus: Priority, stop bits, baudrate *	16		19200/even/1stop
										Even parity, 1 stop bit (default)			
										Odd parity, 1 stop bit			
										No parity, 2 stop bits			
										No parity, 1 stop bit			
			513	40514	0					0 1 0 1			
			513	40514	1					0 0 1 1			
										1200bit/s			
										2400bit/s			
										4800bit/s			
										9600bit/s			
										19200bit/s (default)			
										38400bit/s			
										57600bit/s			
										76800bit/s			
			513	40514	2					0 1 0 1 0 1 0 1 0			
			513	40514	3					0 0 1 1 0 0 1 1 0			
			513	40514	4					0 0 0 0 1 1 1 1 0			
			513	40514	5					0 0 0 0 0 0 0 0 1			
			513	40514	6					reserved, set to 0			
			513	40514	7					reserved, set to 0			
%	%	%	514	40515		%	%	%	%	Modbus: Modus: 0=RTU, 1=ASCII *	0	RTU	Modus: RTU
%	%	%	515	40516		%	%	%	%	Modbus: Timeout in ms	7D0	2000	Timeout: 2000ms

2. Master Controller ==>>> Compressor

Table 2.1 : Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC BDRcv	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
xBS0	1.0	0	0	40001	0	0	0	0	1.0	SAM connection	yes	no	Duration
xBS1	1.1	0	0	40001	1	0	1	0	1.1				
xBS2	1.2	0	0	40001	2	0	2	0	1.2				
xBS3	1.3	0	0	40001	3	0	3	0	1.3				
xBS4	1.4	0	0	40001	4	0	4	0	1.4				
xBS5	1.5	0	0	40001	5	0	5	0	1.5				
xBS6	1.6	0	0	40001	6	0	6	0	1.6				
xBS7	1.7	0	0	40001	7	0	7	0	1.7				
xBS8	0.0	0	0	40001	8	1	0	0	0.0				
xBS9	0.1	0	0	40001	9	1	1	0	0.1				
xBS10	0.2	0	0	40001	10	1	2	0	0.2				
xBS11	0.3	0	0	40001	11	1	3	0	0.3				
xBS12	0.4	0	0	40001	12	1	4	0	0.4				
xBS13	0.5	0	0	40001	13	1	5	0	0.5				
xBS14	0.6	0	0	40001	14	1	6	0	0.6				
xBS15	0.7	0	0	40001	15	1	7	0	0.7				

- 1) set 1 bit only
- 2) set 1 bit only
- 3) set 1 bit only
- 4) set 1 bit only

2. Master Controller ==>>> Compressor

Table 2.1 : Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

1) set 1 bit only

2) set 1 bit only

3) set 1 bit only

4) set 1 bit only

JSON-RPC BDRcv	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
xBS16	3.0	1	1	40002	0	2	0	1	3.0	Mode compressor ON from "I" key only	yes ¹⁾	local	duration
xBS17	3.1	1	1	40002	1	2	1	1	3.1	Mode compressor ON from "I" key and clock	yes ¹⁾	local	duration
xBS18	3.2	1	1	40002	2	2	2	1	3.2				
xBS19	3.3	1	1	40002	3	2	3	1	3.3				
xBS20	3.4	1	1	40002	4	2	4	1	3.4	Mode compressor ON from "I" key and external contact	yes ¹⁾	local	duration
xBS21	3.5	1	1	40002	5	2	5	1	3.5	Mode compressor ON from "I" key and clock or external contact	yes ¹⁾	local	duration
xBS22	3.6	1	1	40002	6	2	6	1	3.6	Mode compressor ON from "I" key and bus master	yes (ON/OFF via xBS23) ¹⁾	local	duration
xBS23	3.7	1	1	40002	7	2	7	1	3.7	Compressor ON / OFF by bus master (if xBS22 = 1)	ON	OFF	duration
xBS24	2.0	1	1	40002	8	3	0	1	2.0	DUAL control mode	DUAL ²⁾	local if xBS24 and xBS25 and xBS27 is "0"	duration
xBS25	2.1	1	1	40002	9	3	1	1	2.1	QUADRO control mode	QUADRO ²⁾		duration
xBS26	2.2	1	1	40002	10	3	2	1	2.2				
xBS27	2.3	1	1	40002	11	3	3	1	2.3	DYNAMIC control mode	DYNAMIC ²⁾		duration
xBS28	2.4	1	1	40002	12	3	4	1	2.4				
xBS29	2.5	1	1	40002	13	3	5	1	2.5				
xBS30	2.6	1	1	40002	14	3	6	1	2.6				
xBS31	2.7	1	1	40002	15	3	7	1	2.7				

2. Master Controller ==>>>> Compressor

Table 2.1 : Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

1) set 1 bit only

2) set 1 bit only

3) set 1 bit only

4) set 1 bit only

JSON-RPC BDRcv	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
xBS32	5.0	2	2	40003	0	4	0	2	5.0				
xBS33	5.1	2	2	40003	1	4	1	2	5.1				
xBS34	5.2	2	2	40003	2	4	2	2	5.2				
xBS35	5.3	2	2	40003	3	4	3	2	5.3				
xBS36	5.4	2	2	40003	4	4	4	2	5.4	Mode load-idle signal from external contact	yes ³⁾	local	duration
xBS37	5.5	2	2	40003	5	4	5	2	5.5				
xBS38	5.6	2	2	40003	6	4	6	2	5.6	Mode load-idle signal from bus master	yes (load/idle via xBS39) ³⁾	local	duration
xBS39	5.7	2	2	40003	7	4	7	2	5.7	Load run by bus master (if xBS38 = 1)	load run	idle	duration
xBS40	4.0	2	2	40003	8	5	0	2	4.0	Mode load-idle signal from pressure regualtor, pA permanent	yes ⁴⁾	local	duration
xBS41	4.1	2	2	40003	9	5	1	2	4.1	Mode load-idle signal from pressure regualtor, pB permanent	yes ⁴⁾	local	duration
xBS42	4.2	2	2	40003	10	5	2	2	4.2	Mode load-idle signal from pressure regualtor, pA/pB change via clock	yes ⁴⁾	local	duration
xBS43	4.3	2	2	40003	11	5	3	2	4.3	Mode load-idle signal from pressure regualtor, pA/pB change via timer	yes ⁴⁾	local	duration
xBS44	4.4	2	2	40003	12	5	4	2	4.4				
xBS45	4.5	2	2	40003	13	5	5	2	4.5	Mode load-idle signal from pressure regualtor, pA/pB change via external contact	yes ⁴⁾	local	duration
xBS46	4.6	2	2	40003	14	5	6	2	4.6	Mode load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet)	yes ⁴⁾	local	duration
xBS47	4.7	2	2	40003	15	5	7	2	4.7				

2. Master Controller ==>>> Compressor

Table 2.1 : Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC BDRcv	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
xBS48	7.0	3	3	40004	0	6	0	3	7.0	Set speed	bus master (iDV4)	local	duration
xBS49	7.1	3	3	40004	1	6	1	3	7.1				
xBS50	7.2	3	3	40004	2	6	2	3	7.2				
xBS51	7.3	3	3	40004	3	6	3	3	7.3				
xBS52	7.4	3	3	40004	4	6	4	3	7.4				
xBS53	7.5	3	3	40004	5	6	5	3	7.5				
xBS54	7.6	3	3	40004	6	6	6	3	7.6	Enable message acknowledgement by bus master	yes (ack. via xBS55)	local	duration
xBS55	7.7	3	3	40004	7	6	7	3	7.7	Acknowledge signal for alarms and warnings (if xBS54=1)	L→H: acknowledge		flank
xBS56	6.0	3	3	40004	8	7	0	3	6.0	Enable compressor autostart by bus master	yes (set via xBS57)	local	duration
xBS57	6.1	3	3	40004	9	7	1	3	6.1	Compressor restart (if xBS56=1)	yes	no	duration
xBS58	6.2	3	3	40004	10	7	2	3	6.2				
xBS59	6.3	3	3	40004	11	7	3	3	6.3				
xBS60	6.4	3	3	40004	12	7	4	3	6.4				
xBS61	6.5	3	3	40004	13	7	5	3	6.5				
xBS62	6.6	3	3	40004	14	7	6	3	6.6				
xBS63	6.7	3	3	40004	15	7	7	3	6.7				

2. Master Controller ==>>> Compressor

Table 2.1 : Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

JSON-RPC BDRcv	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
xBS64	35.0	17	17	40018	0	34	0	17	35.0				
xBS65	35.1	17	17	40018	1	34	1	17	35.1				
xBS66	35.2	17	17	40018	2	34	2	17	35.2				
xBS67	35.3	17	17	40018	3	34	3	17	35.3				
xBS68	35.4	17	17	40018	4	34	4	17	35.4	Set pressure pA (switching point and switching difference) via	bus master (iDV0+iDV1)	local	duration
xBS69	35.5	17	17	40018	5	34	5	17	35.5				
xBS70	35.6	17	17	40018	6	34	6	17	35.6	Set pressure pB (switching point and switching difference) via	bus master (iDV2+iDV3)	local	duration
xBS71	35.7	17	17	40018	7	34	7	17	35.7				
xBS72	34.0	17	17	40018	8	35	0	17	34.0				
xBS73	34.1	17	17	40018	9	35	1	17	34.1				
xBS74	34.2	17	17	40018	10	35	2	17	34.2				
xBS75	34.3	17	17	40018	11	35	3	17	34.3				
xBS76	34.4	17	17	40018	12	35	4	17	34.4				
xBS77	34.5	17	17	40018	13	35	5	17	34.5	Get actual air receiver system pressure via	bus master (iDV9)	local	duration
xBS78	34.6	17	17	40018	14	35	6	17	34.6				
xBS79	34.7	17	17	40018	15	35	7	17	34.7				

- 1) set 1 bit only
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2. Master Controller ==>>> Compressor

Table 2.1 : Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 2

- 1) set 1 bit only
 2) set 1 bit only
 3) set 1 bit only
 4) set 1 bit only

JSON-RPC BDRcv	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet EtherNet/IP		SAM Send/Receive		Description	Signal		
	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind
xBS80	37.0	18	18	40019	0	36	0	18	37.0				
xBS81	37.1	18	18	40019	1	36	1	18	37.1				
xBS82	37.2	18	18	40019	2	36	2	18	37.2	Heat request (valve V10 and/or valve V14) (if xBS83 = 1 or xBS85 = 1)	yes	local	duration
xBS83	37.3	18	18	40019	3	36	3	18	37.3	Enable heat request valve V10 by bus master	yes (set via xBS82)	local	duration
xBS84	37.4	18	18	40019	4	36	4	18	37.4				
xBS85	37.5	18	18	40019	5	36	5	18	37.5	Enable heat request valve V14 by bus master	yes (set via xBS82)	local	duration
xBS86	37.6	18	18	40019	6	36	6	18	37.6	Water system: Get actual setpoint temperature control T14 via	bus master (iDV14)	local	duration
xBS87	37.7	18	18	40019	7	36	7	18	37.7				
xBS88	36.0	18	18	40019	8	37	0	18	36.0				
xBS89	36.1	18	18	40019	9	37	1	18	36.1				
xBS90	36.2	18	18	40019	10	37	2	18	36.2				
xBS91	36.3	18	18	40019	11	37	3	18	36.3				
xBS92	36.4	18	18	40019	12	37	4	18	36.4				
xBS93	36.5	18	18	40019	13	37	5	18	36.5				
xBS94	36.6	18	18	40019	14	37	6	18	36.6				
xBS95	36.7	18	18	40019	15	37	7	18	36.7				